

AEGIS: AI Emergency & Geospatial Intelligence System

AI-01 — AI Disaster Response Intelligence Platform

THE CATALYST

HACKATHON MVP

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Understanding the Ultimate Goal

Disasters — earthquakes, floods, droughts, nuclear incidents, chemical runaways — generate overwhelming volumes of data. Human operators alone cannot process it in time. AEGIS is designed to close that gap.

Collect

Aggregate satellite imagery, emergency calls, official reports, and social media simultaneously

Analyze

Process situational data in real time to build a unified operational picture

Prioritize

Identify high-urgency zones and optimize displacement of rescue resources

Recommend

Surface actionable response guidance and safe evacuation corridors on a live RAG-integrated dashboard

Understanding Current Gaps Through Case Studies

KEDARNATH FLOODS — 2013

A cloudburst triggered flash floods that killed **over 5,700 people**. A tragedy that could have been avoided if weather reports were actually reaching people and rescue groups before it was too late.

If a system existed in the Kedarnath Valley that could say "23 people spotted, 8 buildings damaged, 2 roads blocked" — nobody would have had to physically go and assess the situation. The AI would have counted all of these things automatically.

CHAMOLI GLACIER BURST — 2021

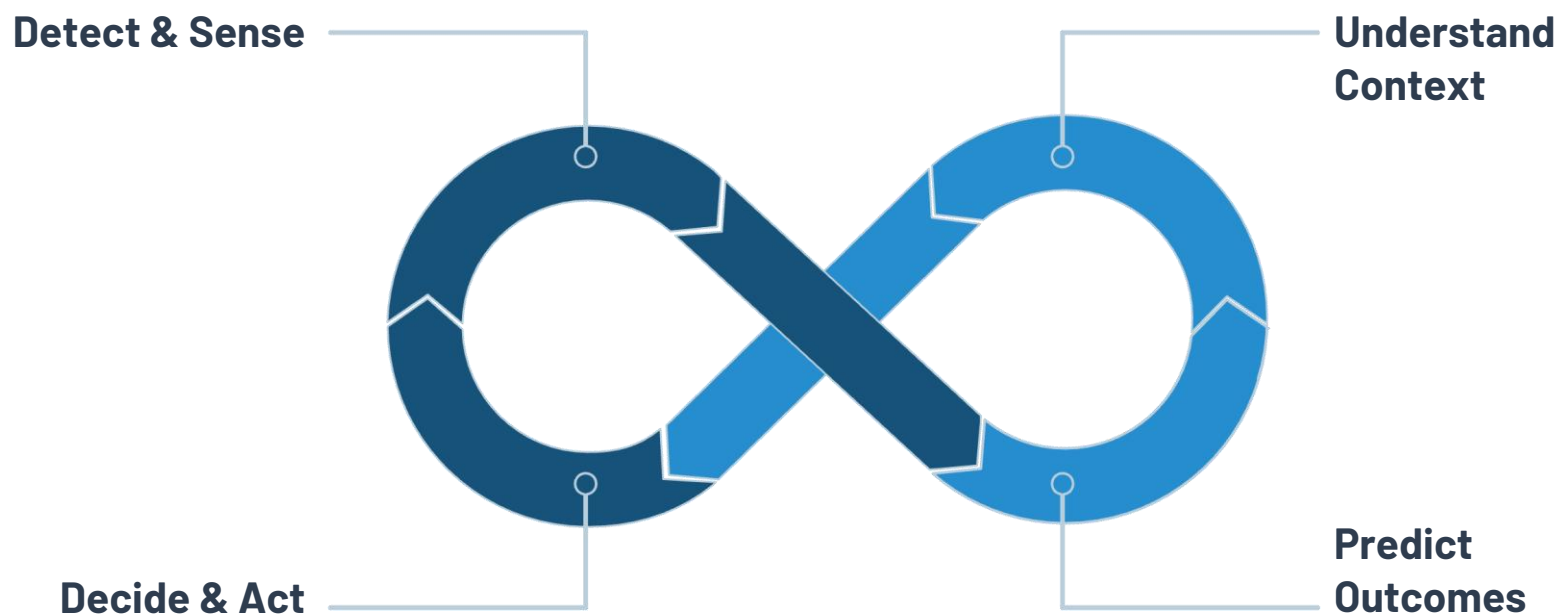
A glacier collapse destroyed two hydropower projects and killed **200+ people**. The communication was as outdated as a manual phone call. When the flood hit, it was too fast to let people escape — even though this exact risk had been flagged 8 years earlier.

The flash flood didn't just hit one point; it spread down the entire valley, destroying 2 power plants. A satellite imaging system showing exactly how far the water and debris had spread would have been far more actionable than ambiguous manual reports.

Core Philosophy: From Situational Awareness to Decisive Action

Most dashboards tell you what is happening. AEGIS tells you what happens next — and what you should do before it does.

AEGIS ingests situational data, predicts escalation trajectories, maps collateral damage, treats silent zones as distress signals, allows responders to simulate interventions before committing resources, and learns from every outcome.



This closed-loop architecture ensures that every incident makes the system more precise — compounding institutional knowledge into faster, more accurate responses over time.

Core Features



Predictive Intelligence & Severity Scoring

Forecasts risk at T+30m, T+60m, and T+180m using rate-of-rise and hydrodynamic models, then rolls population exposure, flood depth, infrastructure risk, and telecom status into one per-zone severity score, so responders can tell at a glance which zone actually needs attention first.



Cascading & Silent Crisis Detection

Tracks how one event snowballs into the next, like a submerged bridge cutting off a hospital, which then threatens a nearby power substation. Also flags zones where zero SOS calls probably means the phones are dead, not that everyone's okay.



SOS Handling & Evidence Verification

Geotags SOS and emergency reports as they come in, then cross-checks them against civilian, sensor, and official sources so nothing gets acted on blindly. Each report gets a Data Trust Index, giving responders a quick sense of how reliable it actually is.



Computer Vision & Mission Optimization

Pulls flood extent and structural damage straight from drone or satellite imagery, then combines that with live ground conditions to pick rescue teams by what they can actually do — boats, medical gear, ability to cross submerged roads — instead of just who's closest on a map.



What-If Simulation & Adaptive Learning

Lets responders test a plan, like evacuating Zone 7 and sending Team R2, in a Monte Carlo sandbox before committing anyone. Afterward, it checks predictions against what really happened and applies small corrections so the next forecast is a bit sharper.

Implementation & Technology Stack

Frontend

- **React** — state management and HUD rendering
- **TypeScript** — typed interfaces for reliability
- **Vite** — optimized build and bundling
- **Tailwind CSS** — responsive UI components
- **React Context API** — lightweight state coordination

Backend

- **Python + FastAPI** — high-performance REST services
- **Uvicorn** — ASGI server for high-throughput HTTP
- **Pytest** — automated test suite for reliability

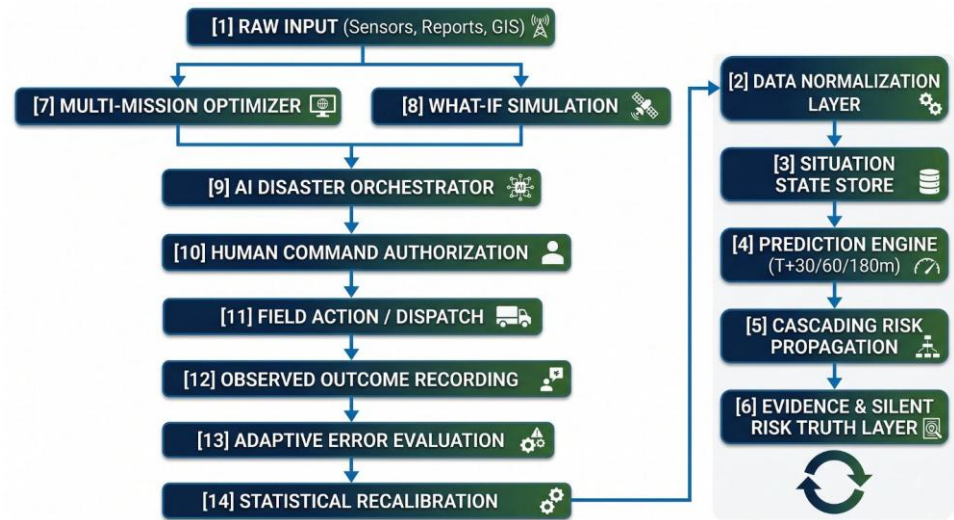
AI Layer

- **Google Gemini 2.5 Flash** — primary LLM for NLP and command briefings
- **Google Gen AI SDK** — official integration layer
- **ReAct Tool Calling** — orchestration pattern that executes deterministic engine queries before response generation, ensuring factual grounding

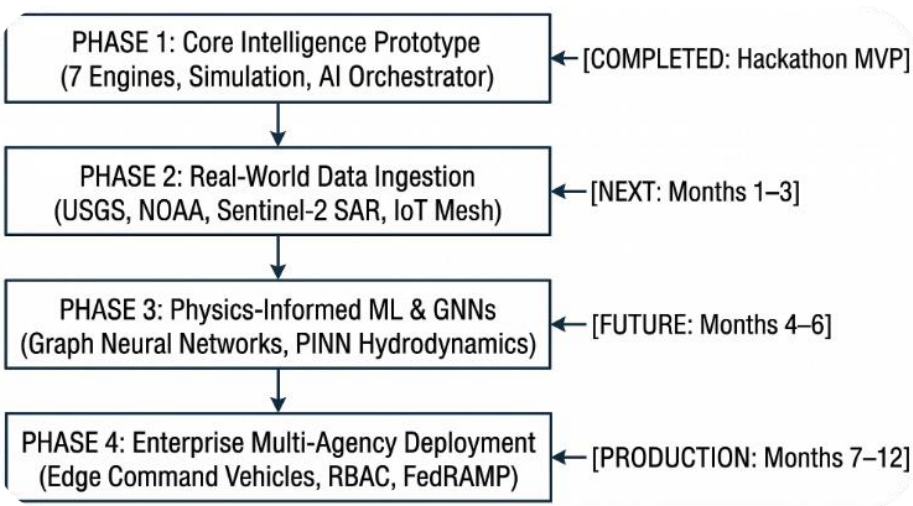
📌 The ReAct pattern prevents hallucination by enforcing structured tool calls against live disaster engines prior to any AI-generated output.

Data Flow & Development Phases

System Data Flow



Phased Deployment Roadmap



1

Phase 1 – MVP

7 engines, simulation, AI orchestrator — **Completed**

2

Phase 2 – Data Ingestion

USGS, NOAA, Sentinel-2 SAR, IoT Mesh — **Months 1-3**

3

Phase 3 – Physics ML

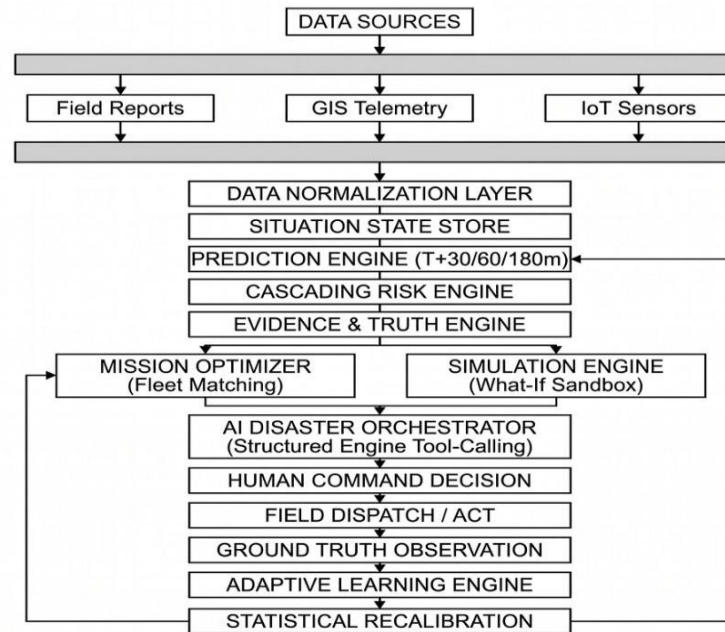
Graph Neural Networks, PINN Hydrodynamics — **Months 4-6**

4

Phase 4 – Enterprise

Edge Command Vehicles, RBAC, FedRAMP — **Months 7-12**

Architectural Flow



Raw inputs from field reports, GIS telemetry, and IoT sensors are normalized and stored in a Situation State Store. The Prediction Engine generates T+30/60/180 minute forecasts, which feed both the Cascading Risk Engine and the Evidence & Truth Layer. The Mission Optimizer and Simulation Engine operate in parallel before routing through the AI Disaster Orchestrator to Human Command Decision — ensuring no autonomous action is taken without authorized human oversight.

Impact: Transforming the First Critical Hour

AEGIS is not autonomous — it is a **decision-support system**. It issues probabilistic scores grounded in visual data and local ground truth, not guarantees. Its value lies in three specific capabilities that existing platforms cannot replicate:

-  **Catching Silent Crises**
Identifies zones that appear safe on paper but exhibit communication breakdown patterns consistent with active distress
-  **Pre-Mission Simulation**
Enables response teams to test intervention plans in a virtual environment before committing personnel and equipment — eliminating costly trial-and-error in the field
-  **Capability-Based Dispatch**
Assigns rescue teams based on skill profile and available resources, not merely proximity or deployment readiness — maximizing the probability of successful outcomes

 These three capabilities alone could measurably compress response latency and improve survival rates in the first critical hour of a large-scale disastrous event